

Generating Particle Properties

In this appendix, we present algorithms for generating particle properties from prescribed distribution functions. Generating particle properties (also referred to as “sampling” particle properties) from a prescribed distribution function is important for two main reasons. First, particles must be generated and/or their properties must be specified at domain boundaries within a direct simulation Monte Carlo calculation, for example at inflow and solid wall boundaries. Second, post-collision properties of particles are often sampled from specific distribution functions, for example the equilibrium distribution functions used in the Borgnakke–Larsen (BL) model. In this appendix, we outline algorithms for generating particles based on equilibrium distributions for a volume of gas, a gas flux across a plane, a gas flux off of a solid surface, and we also outline the acceptance–rejection algorithm that is generally applicable to any distribution function.

All algorithms require the availability of successive random numbers, R , that are uniformly distributed over the range, $0 \leq R \leq 1$. In addition, a normalized distribution function (f_x) must be specified,

$$\int_a^b f_x dx = 1, \quad a \leq x \leq b \quad (\text{A.1})$$

such that $f_x dx$ represents the probability that the value of x lies between x and $x + dx$.

A.1 Cumulative Distribution Function

In some cases, the cumulative distribution function can be used to obtain an expression for generating samples (particle properties) directly from a distribution function. The cumulative distribution function (F_x) is defined as

$$F_x \equiv \int_a^x f_x dx, \quad \text{and therefore, } 0 \leq F_x \leq 1 \quad (\text{A.2})$$

If the distribution function, f_x , is of a certain form, it may be possible to invert Eq. A.2 and solve for x . In this case, sampling the function F_x provides a means to sample values of x that follow the distribution f_x .

A.1.1 Generating Random Particle Positions

As an example, consider the trivial problem of generating a random x -position for a particle, where $a \leq x \leq b$. All x -positions in this range are equally likely and therefore $f_x = \text{Constant}$. From Eq. A.1,

$$f_x = \frac{1}{b-a} \quad (\text{A.3})$$

and therefore,

$$F_x = \int_a^x f_x dx = \int_a^x \frac{1}{b-a} dx = \frac{x-a}{b-a} \quad (\text{A.4})$$

Since $0 \leq F_x \leq 1$, we can sample a random value of F_x by drawing a random number ($0 \leq R \leq 1$), and selecting $F_x = R$. Using Eq. A.4, the value of x that corresponds to R is then

$$x = a + R(b-a) \quad (\text{A.5})$$

This is the expected result for the random selection of an x -position between a and b . The same result (Eq. A.5) can be used to generate random x , y , and z positions within specified cell or domain boundaries.

A.1.2 Generating Radial Positions for Axisymmetric Flow

As another example, consider generating a random radial position (r) of a particle within a uniform density axisymmetric flow, where $a \leq r \leq b$. In this case, since the cross-sectional area is proportional to r , the probability of finding a particle with a specific value of r is also proportional to r . Now, $f_r = \text{Constant} \times r$, and from Eq. A.1 we find

$$f_r = \frac{2r}{b^2 - a^2} \quad (\text{A.6})$$

and therefore,

$$F_r = \int_a^r f_r dr = \frac{r^2 - a^2}{b^2 - a^2} = R \quad (\text{A.7})$$

The final expression to generate a random value of r , for this particular case, becomes

$$r = \sqrt{a^2 + R(b^2 - a^2)} \quad (\text{A.8})$$

A.1.3 Generating Particle Velocities from a Maxwell–Boltzmann Distribution

In a typical DSMC calculation, particle inflow boundaries are located in regions where the gas is in a known equilibrium state and the velocity distribution function is a Maxwell–Boltzmann distribution. For this reason, an efficient algorithm for sampling a Maxwell–Boltzmann distribution is desired.

Consider the Maxwell–Boltzmann distribution function for the thermal velocity of a particle in the i -coordinate direction ($-\infty \leq C'_i \leq +\infty$),

$$f_0(C'_i) = \frac{\beta}{\sqrt{\pi}} e^{-\beta^2 C_i'^2}, \quad \text{where } \beta = \sqrt{\frac{m}{2kT_{tr}}} \quad (\text{A.9})$$

The cumulative distribution function is therefore

$$F(C'_i) = \frac{\beta}{\sqrt{\pi}} \int_{-\infty}^{C'_i} e^{-\beta^2 C_i'^2} dC'_i = \frac{1}{2} [1 + \operatorname{erf}(\beta C'_i)] = R \quad (\text{A.10})$$

Here, $\operatorname{erf}(x)$ represents the error function (refer to Appendix E) and there is no analytical solution for C'_i in terms of R . For such cases, where the corresponding cumulative distribution function cannot be inverted, the acceptance–rejection procedure described later in Section A.2, can be used to sample properties from any distribution function.

However, for the particular case of a Maxwell–Boltzmann velocity distribution, Eq. A.9 can be mathematically reformulated such that the resulting cumulative distribution function can be inverted. To reformulate the problem, consider the joint probability of finding *two* thermal velocity values (C'_i and Z'_i) in the i -coordinate direction. Specifically, consider the probability of finding C'_i between C'_i and $C'_i + dC'_i$ in addition to Z'_i between Z'_i and $Z'_i + dZ'_i$. The joint probability, corresponding to a Maxwell–Boltzmann distribution is

$$\begin{aligned} f_0(C'_i) dC'_i f_0(Z'_i) dZ'_i &= \frac{\beta}{\sqrt{\pi}} e^{-\beta^2 C_i'^2} dC'_i \frac{\beta}{\sqrt{\pi}} e^{-\beta^2 Z_i'^2} dZ'_i \\ &= \frac{\beta^2}{\pi} e^{-\beta^2 (C_i'^2 + Z_i'^2)} dC'_i dZ'_i. \end{aligned} \quad (\text{A.11})$$

This expression can be converted to cylindrical polar coordinates using the following transformation:

$$\begin{aligned} C'_i &= r \cos \theta \\ Z'_i &= r \sin \theta \end{aligned} \quad (\text{A.12})$$

where

$$\frac{\partial (C'_i, Z'_i)}{\partial (r, \theta)} = \begin{vmatrix} (\partial C'_i / \partial r) & (\partial C'_i / \partial \theta) \\ (\partial Z'_i / \partial r) & (\partial Z'_i / \partial \theta) \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \quad (\text{A.13})$$

and therefore, $dC'_i dZ'_i = r dr d\theta$

In cylindrical polar coordinates, Eq. A.11 becomes

$$\begin{aligned} f_0(C'_i) dC'_i f_0(Z'_i) dZ'_i &= \frac{\beta^2}{\pi} e^{-\beta^2 r^2} r dr d\theta \\ &= \underbrace{e^{-\beta^2 r^2} d(\beta^2 r^2)}_{f(\beta^2 r^2) d(\beta^2 r^2)} \times \underbrace{\frac{d\theta}{2\pi}}_{f(\theta) d\theta} \end{aligned} \quad (\text{A.14})$$

and the cumulative distribution function can now be written as

$$F(C'_i, Z'_i) = F(\beta^2 r^2) F(\theta) \quad (\text{A.15})$$

Each separate cumulative distribution can be inverted as follows:

$$F(\theta) = \int_0^\theta \frac{d\theta}{2\pi} = \frac{\theta}{2\pi} = R_1 \quad (\text{A.16})$$

which leads to

$$\theta = 2\pi R_1 \quad (\text{A.17})$$

Furthermore,

$$F(\beta^2 r^2) = \int_0^{\beta^2 r^2} e^{-\beta^2 r^2} d(\beta^2 r^2) = 1 - e^{-\beta^2 r^2} = R_2 \quad (\text{A.18})$$

and since generating a random value of $1 - R_2$ is the same as generating a random value R_2 , we have

$$r = \frac{\sqrt{-\ln(R_2)}}{\beta} \quad (\text{A.19})$$

In this manner, by drawing two independent random numbers R_1 and R_2 , the simple expressions given by Eqs. A.17 and A.19 can be used to determine r and θ , and therefore determine two thermal velocities, C'_i and Z'_i using Eq. A.12.

Certain DSMC algorithms often require the generation of a relatively small number of particles, for example, particles within a single flow cell or fluxing through a single cell face. Although the use of Eq. A.12 to generate both C'_i and Z'_i values will indeed result in particle velocities following the desired Maxwell–Boltzmann distribution, the values of C'_i and Z'_i are correlated. To avoid such correlations, it is typical to generate only one velocity value and disregard the other. In this manner, particle velocity vectors (C_x, C_y, C_z) can be sampled from a specified Maxwell–Boltzmann distribution (corresponding to bulk velocity components $\langle C_x \rangle$, $\langle C_y \rangle$, $\langle C_z \rangle$, and translational temperature T_{tr}), using the following expression:

$$C_i = \langle C_i \rangle + C'_i = \langle C_i \rangle + \sqrt{\frac{2kT_{tr}}{m}} \sin(2\pi R_1) \sqrt{-\ln(R_2)} \quad (\text{A.20})$$

Here $i = x, y$, or z and $R_1 \neq R_2$. Therefore, using Eq. A.20, two independent random numbers are required to generate each component (x, y, z) of

a particle velocity vector from a Maxwell–Boltzmann distribution. This simple algorithm is much more efficient than the general acceptance–rejection algorithm discussed later in Section A.2.

A.1.4 Generating Particle Internal Energies from a Boltzmann Distribution

As discussed in Chapter 6 (Section 6.4.2, Eq. 6.64), the energy distribution function for any continuous energy mode (ϵ_i) in an equilibrium gas, with corresponding ζ_i degrees of freedom, can be expressed by

$$f\left(\frac{\epsilon_i}{kT_i}, \zeta_i\right) d\left(\frac{\epsilon_i}{kT_i}\right) = \frac{1}{\Gamma(\zeta_i/2)} \left(\frac{\epsilon_i}{kT_i}\right)^{\zeta_i/2-1} e^{-\epsilon_i/kT_i} d\left(\frac{\epsilon_i}{kT_i}\right) \quad (\text{A.21})$$

For the special case of $\zeta_i = 2$, corresponding to the rotational energy mode for a diatomic species and also corresponding to the vibrational mode (if vibration is fully excited), this expression reduces to

$$f\left(\frac{\epsilon_i}{kT_i}\right) d\left(\frac{\epsilon_i}{kT_i}\right) = e^{-\epsilon_i/kT_i} d\left(\frac{\epsilon_i}{kT_i}\right) \quad (\text{A.22})$$

This is precisely the same form as the first term in Eq. A.14, and therefore the cumulative distribution function has the same form as Eq. A.18. This can be solved for a value of the internal energy,

$$\epsilon_i = -\ln(R)kT_i \quad (\text{A.23})$$

which is the analogue of Eq. A.19. Here, T_i is the equilibrium temperature of the internal energy mode (i). In general, one may wish to generate particles corresponding to a state of thermal nonequilibrium where translational, rotational, and vibrational temperatures are different ($T_{\text{tr}} \neq T_{\text{rot}} \neq T_{\text{vib}}$), yet each energy mode (i) is characterized by a Maxwell–Boltzmann distribution based on T_i .

Quantized vibrational energies may be generated in a number of ways. For example, according to the harmonic oscillator vibrational energy model, the expression to sample vibrational energy is analogous to Eq. A.23, however, using the following truncation:

$$\begin{aligned} \epsilon_{\text{vib}} &= k\theta_{\text{vib}}i_{\text{vib}} \\ i_{\text{vib}} &= \text{floor}\left[-\ln(R) \times \left(\frac{T_{\text{vib}}}{\theta_{\text{vib}}}\right)\right] \end{aligned} \quad (\text{A.24})$$

where i_{vib} is an integer corresponding to the quantized vibrational energy level and θ_{vib} is the characteristic temperature for vibration. Although Eq. A.24 simply rounds the energy down, different binning strategies could be used. Also, the expression could be generalized to an anharmonic oscillator description, or any vibrational energy model for that matter, using the partition function corresponding to the model.

In summary, particle velocity vectors can be generated using Eq. A.20 and particle rotational and vibrational (continuous) energies can be generated using Eq. A.23, according to Maxwell–Boltzmann distribution functions. To generate quantized vibrational energies, Eq. A.24 can be used. To generate particle properties from a general distribution function, the acceptance–rejection technique is now described.

A.2 Acceptance–Rejection Sampling

To sample any continuous distribution function, $f(x)$, the acceptance–rejection algorithm can be employed. The standard inequality used in most DSMC implementations is $u \leq f(y)/M$. Here, the value y is drawn from a uniform distribution and $f(y)$ is evaluated and normalized by a constant that is greater than or equal to the maximum value of the distribution function, $M \geq f(x)|_{\max}$. A second value, u , is then drawn from a uniform distribution, and if the inequality holds, then the value y is accepted, and a sample value is determined as $x = y$. If the inequality does not hold, the value y is rejected and the process is repeated. In this manner, the resulting set of x values follow $f(x)$.

For completeness, we outline the algorithm in the text that follows. To sample a random variable X from a continuous distribution $f(X)$, and obtain a value x for X , the algorithm is as follows:

- (1) Generate a value y for a random variable Y having a distribution $g(Y)$.
- (2) Generate a value u for a random variable U that is independent of Y .
- (3) Choose a constant M , and check if

$$u \leq f(y)/[Mg(y)]$$

- (a) If the preceding expression holds, then accept the generated value y , by setting $x = y$.
- (b) If the preceding expression does not hold, then reject the value y , go back to (1) and regenerate another value of y .

Using the aforementioned algorithm to generate a series of x values, the probability distribution function $f(x)$ is correctly sampled. When using the acceptance–rejection method within a DSMC calculation, the random variable Y is typically set using a uniform distribution, $g(Y) = 1$. Similarly, the random variable U is typically set using another uniform distribution in the interval $(0, 1)$. As mentioned previously, M is set as a value equal to or greater than the maximum value of the distribution function, i.e., $M \geq f(x)|_{\max}$. With these typical settings for DSMC calculations, the inequality used in the acceptance–rejection algorithm simplifies to $u \leq f(y)/M$.

The acceptance–rejection algorithm is also described in detail for post-collision sampling of continuous internal energy in Chapter 6, Section 6.4.4, for sampling quantized vibrational energy in Chapter 7, Section 7.3.3, and also for sampling the properties of particles fluxing across a plane, in Section A.3 in the text that follows. In some cases, it is desirable to sample from a Chapman–Enskog velocity distribution (Eqs. 5.67 and 5.68) instead of a Maxwell–Boltzmann distribution. An efficient algorithm for sampling the Chapman–Enskog distribution has been developed by Garcia and Alder (1998). Finally, algorithms for sampling the generalized Chapman–Enskog distribution, for multispecies mixtures, have been developed by Stephani, Goldstein and Varghese (2013).

A.3 Generation of Particle Flux Across a Planar Element

Often, the computational domain boundaries of a DSMC calculation are represented by a large number of planar elements, such as a subset of cell faces or sometimes a separate triangulated surface. For inflow surfaces, the flux of simulation particles through each planar element and the properties of the particles may be specified at each simulation timestep.

Consider a planar element of area A and with a unit normal vector in Cartesian coordinates $\hat{n} = (n_x\bar{x}, n_y\bar{y}, n_z\bar{z})$ directed *into* the computational domain. It is convenient to carry out all operations in face-normal coordinates $(\bar{x}_f, \bar{y}_f, \bar{z}_f)$ as depicted in Fig. A.1, where the $\bar{x}_f$ direction is aligned with $\hat{n}$, and the $\bar{y}_f$ and $\bar{z}_f$ directions are normal to $\hat{n}$.

First, the dot product between the bulk velocity vector,

$$\bar{V} = (\langle C_x \rangle \bar{x}, \langle C_y \rangle \bar{y}, \langle C_z \rangle \bar{z}) \quad (\text{A.25})$$

and the face unit normal vector is used to define a speed ratio (s_n) in the face-normal direction,

$$s_n \equiv \frac{\bar{V} \cdot \hat{n}}{C_m} \quad (\text{A.26})$$

where $C_m = \sqrt{2kT_{tr}/m} = 1/\beta$ is the mean thermal speed. Rewriting the expression for free molecular mass flux across a plane (Eq. 1.154 derived in Chapter 1 with s_3 now replaced by s_n) we can write the molecular flux as

$$F_n = \frac{nC_m}{2\sqrt{\pi}} \left\{ e^{-s_n^2} + \sqrt{\pi}s_n [1 + \text{erf}(s_n)] \right\} \quad (\text{A.27})$$

where n is the number density. Therefore, F_n represents the local flux of molecules per square meter per second in the $\hat{n}$ direction. The flux of simulated particles is then given by

$$F_{\text{DSMC}} = \frac{F_n A \Delta t}{W_p} \quad (\text{A.28})$$

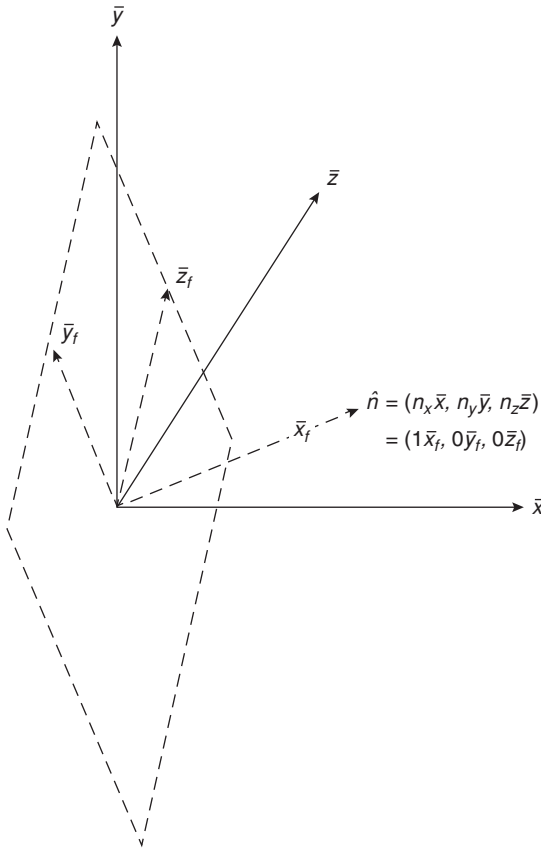


Figure A.1

Face-normal coordinate system used to determine particle flux through a planar element and particle properties.

where A is the area of the element under consideration, Δt is the DSMC timestep, and W_p is the particle weight. Finally, the number of particles generated to simulate the flux through the element is

$$N_{\text{gen}} = \text{floor}[F_{\text{DSMC}} + R], \quad \text{where } 0 \leq R \leq 1 \text{ is a random number,} \quad (\text{A.29})$$

since only an integer number of particles can be generated at each timestep.

The properties of each of these N_{gen} particles are then sampled from equilibrium distribution functions. The thermal velocity components in the $\bar{y}_f$ and $\bar{z}_f$ directions (normal to $\hat{n}$) are sampled from Maxwell–Boltzmann distribution functions using Eq. A.20 derived previously in Section A.1.3. Specifically,

$$\begin{aligned} C'_{y_f} &= C_m \sin(2\pi R_1) \sqrt{-\ln(R_2)} \\ C'_{z_f} &= C_m \sin(2\pi R_3) \sqrt{-\ln(R_4)} \end{aligned} \quad (\text{A.30})$$

Note here that $\langle C_{y_f} \rangle$ and $\langle C_{z_f} \rangle$ were set to zero, so that only the thermal velocity components (C'_{y_f} and C'_{z_f}) are calculated. The bulk velocity will be added to the thermal velocities in a later step, after switching back to the Cartesian coordinate system.

The particle velocity component in the $\bar{x}_f$ direction (i.e., the $\hat{n}$ direction) requires special attention. Specifically, only positive values of this velocity component should be sampled. This involves sampling from a partial, and biased, Maxwell–Boltzmann distribution. The relevant distribution function was previously derived in Chapter 1 to determine the incident flux of a particle property to a planar surface element in free molecular flow, and was given in Eq. 1.147. Referring to Eq. 1.147 and related discussion, the appropriate distribution function for the thermal velocity component in the $\hat{n}$ direction (C'_{x_f}) is identical to Eq. 1.147, where $s_3 = s_n$, $C'_3 = C'_{x_f}$, and $Q = 1$:

$$\begin{aligned} & \int_{C'_{x_f} = -s_n/\beta}^{\infty} f(C'_{x_f}) dC'_{x_f} \\ &= \int_{C'_{x_f} = -s_n/\beta}^{\infty} \frac{1}{\sqrt{\pi}} \left(\beta C'_{x_f} + s_n \right) e^{-\beta^2 C'^2_{x_f}} dC'_{x_f} \end{aligned} \quad (\text{A.31})$$

In this manner, particles with thermal velocities less than $-s_n/\beta$ in the $\hat{n}$ direction will not flux through the planar surface into the simulation domain.

To sample values of C'_{x_f} from this distribution function, the acceptance–rejection algorithm (outlined in Section A.2) is used. This procedure requires a maximum value of the distribution function, $M = f(C'_{x_f})|_{\max}$, and then determining the ratio of $f(C'_{x_f})/M$. The maximum of (C'_{x_f}) occurs for

$$\beta C'_{x_f} = \left\{ \sqrt{s_n^2 + 2} - s_n \right\} / 2 \quad (\text{A.32})$$

and therefore,

$$\frac{f(y)}{M} = \frac{f(y)}{f(y)|_{\max}} = K(y + s_n)e^{-y^2} \quad (\text{A.33})$$

where

$$\begin{aligned} K &= \frac{2}{s_n + h} \exp[1/2 + s_n(s_n - h)/2] \\ h &= \sqrt{s_n^2 + 2} \end{aligned} \quad (\text{A.34})$$

Here, $y = \beta C'_{x_f}$ is a random variate representing a trial value for $\beta C'_{x_f}$. To ensure sampling of the distribution tails, one might calculate a random value for C'_{x_f} in the range $-3C_m \leq C'_{x_f} \leq +3C_m$, or equivalently, $-3 \leq y \leq 3$. For velocities outside of this range, the probability density would be low (i.e., $< e^{-3^2}$), and therefore such velocities would be quite rare and could be neglected. In this case, the acceptance–rejection algorithm would consist of the following steps:

- (1) Calculate the value, $y = -3 + 6R_1$, where $0 \leq R_1 \leq 1$ is a random number.
- (2) Generate a second random number, $0 \leq R_2 \leq 1$.

(3) Calculate $f(y)/M$ using Eq. A.33 and check if

$$R_2 \leq f(y)/M$$

- (a) If the preceding expression holds, then accept the generated value y , by setting $C'_{x_f} = y/\beta$.
- (b) If the preceding expression does not hold, then reject the value y , go back to (1), and generate another value of y .

In summary, for each of the N_{gen} particles (Eq. A.29), thermal velocity components C'_{y_f} , and C'_{z_f} are generated using Eq. A.30, and component C'_{x_f} is generated using Eq. A.33 within the acceptance–rejection algorithm described earlier. These values represent thermal velocity components in the $(\bar{x}_f, \bar{y}_f, \bar{z}_f)$ coordinate frame of reference. Using the face-element normal vector, $\hat{n} = (n_x\bar{x}, n_y\bar{y}, n_z\bar{z})$, these velocity components can be transformed back into the Cartesian coordinate frame of reference, yielding thermal velocities C'_x , C'_y , and C'_z . Finally, the full particle velocity vector is obtained by adding the specified bulk flow velocity as follows:

$$\begin{aligned} C_x &= \langle C_x \rangle + C'_x \\ C_y &= \langle C_y \rangle + C'_y \\ C_z &= \langle C_z \rangle + C'_z \end{aligned} \tag{A.35}$$

At this stage, the velocity vectors for all N_{gen} particles have been assigned in Cartesian coordinates. The internal energies of each of the N_{gen} particles are obtained by sampling the appropriate Maxwell–Boltzmann energy distribution function, using Eq. A.23 for continuous energy distributions, or Eq. A.24 in the case of a quantized vibrational energy distribution.

A.4 Generating Particle Properties Resulting from Surface Collisions

When a particle collides with a solid surface boundary in a DSMC simulation, the post-collision particle properties must be updated to achieve the desired gas–surface boundary condition. The standard boundary condition used in DSMC for collisions with solid surfaces is diffuse reflection and full thermal accommodation. This means that the particles leave the surface according to a Maxwell–Boltzmann distribution corresponding to zero bulk velocity and the surface temperature (T_{surface}).

For most materials, below the scale of a surface element (which in DSMC is often greater than or equal to the local mean free path) lies complex surface structure at the micro- to nanoscale. Gas particles may undergo many surface collisions before leaving the microstructure and returning to the gas flow. For this reason, specifying diffuse reflection relative to the planar surface element, and full thermal accommodation to the surface temperature, are excellent assumptions.

When generating particle velocities and internal energies resulting from a surface collision, it is convenient to work in face-normal coordinates (as shown in Fig. A.1). In fact, the procedure is analogous to that outlined in Section A.3 above, except that the bulk velocity in the $\hat{n}$ direction is now zero (i.e., $s_n = 0$). As a result, the distribution function for the thermal velocity component in the $\hat{n}$ direction is obtained from Eq. A.31 with $s_n = 0$, and a different normalization factor resulting from restriction that $0 \leq C'_{x_f} \leq \infty$:

$$f(C'_{x_f}) dC'_{x_f} = 2\beta^2 C'_{x_f} e^{-\beta^2 C'^2_{x_f}} dC'_{x_f} \quad (\text{A.36})$$

Therefore, the distribution for $\beta^2 C'_{x_f}$ can be written as

$$f(\beta^2 C'_{x_f}) d(\beta^2 C'_{x_f}) = e^{-\beta^2 C'^2_{x_f}} d(\beta^2 C'_{x_f}) \quad (\text{A.37})$$

A distribution function of this form leads to the identical cumulative distribution function analyzed earlier in Eq. A.18, and therefore C'_{x_f} can be obtained using a single random number, $0 \leq R \leq 1$,

$$C'_{x_f} = \frac{\sqrt{-\ln(R)}}{\beta} = C_m \sqrt{-\ln(R)} \quad (\text{A.38})$$

The other thermal velocity components, C'_{y_f} and C'_{z_f} (normal to $\hat{n}$), are calculated using Eq. A.30 as described in Section A.3. Finally, these velocity components must be transformed from the face-normal coordinate system back to the Cartesian coordinate system (using the surface normal vector, $\hat{n}$), to obtain the final thermal velocity components C'_x , C'_y , and C'_z . The internal energies of the particle are obtained by sampling the appropriate Maxwell–Boltzmann energy distribution function, using Eq. A.23 for continuous energy distributions, or Eq. A.24 in the case of a quantized vibrational energy distribution. For full thermal accommodation, all temperatures should correspond to the surface temperature ($T_{\text{tr}} = T_{\text{rot}} = T_{\text{vib}} = T_{\text{surface}}$).

It is important to note that even with these assumptions, DSMC calculations naturally predict “velocity slip” and “temperature jump” phenomena. These are nonequilibrium phenomena where the average velocity of gas molecules at the surface is nonzero, and the average thermal energy of gas molecules at the surface is not the same as that corresponding to the surface temperature. This occurs due to an insufficient number of gas–surface collisions resulting from a combination of low density and/or small length scales for the gas–surface interaction. For example, “velocity slip” and “temperature jump” may be noticeable for flow over a sharp leading edge. In this situation only a fraction of the molecules near the surface have collided with (and accommodated to) the surface, while the remainder have not. This results in local non-Maxwell–Boltzmann velocity distributions with nonzero average velocity and a temperature not equal to the surface temperature.

Since the DSMC method simulates gas–surface interactions on a per-collision basis, there is considerable flexibility to incorporate advanced physical models. Many gas–surface collision models, including for gas–surface

chemical reactions, have been proposed in the literature for a wide range of applications.

A.5 Subsonic and Fluctuating Boundary Conditions

Subsonic DSMC boundary conditions involve generating particles (or updating their properties) at boundaries and, in general, involve the same algorithms outlined in this appendix. However, the precise gas state at a subsonic boundary is not known a priori, rather it is now part of the solution. There are a number of strategies available to implement subsonic boundary conditions in DSMC, some of which are analyzed in a recent study by Farbar and Boyd (2014). The most suitable implementation for subsonic boundary conditions is often problem specific. Whatever strategy is chosen, care should be taken to properly verify the physical accuracy of the subsonic flow result for the problem of interest.

When simulating nonequilibrium flow for microscale applications, not only is the flow typically subsonic, but it can also involve real fluctuations (refer to Fig. 6.1). For such flows, the algorithms presented in this section require modification. In general, if real fluctuations are present within the gas, then such fluctuations should also be prescribed at simulation boundaries. Owing to correlations between hydrodynamic variables, care must be taken when evaluating means and variances, and particle generation (boundary conditions) must also be performed in a consistent manner. Effects of fluctuations and correlations, as well as numerical strategies to model these effects in a physically accurate manner have been described in a series of articles by Tysanner and Garcia (2004, 2005) and Garcia (2007).